

## What `method = "gwet"` Means in `misskappa`

This note documents what the `"gwet"` method does in the current `misskappa` implementation (as opposed to the descriptive label in the manuscript).

### Setup

Let  $X_{ij}$  be the rating for subject  $i \in \{1, \dots, n\}$  by rater  $j \in \{1, \dots, R\}$ , and let  $M_{ij} = 1[X_{ij} \text{ is observed}]$  indicate non-missingness.

Define the estimated marginal and pairwise observation rates

$$\hat{\pi}_j = \frac{1}{n} \sum_{i=1}^n M_{ij}, \quad \hat{\pi}_{jk} = \frac{1}{n} \sum_{i=1}^n M_{ij} M_{ik}.$$

Let  $\ell(\cdot, \cdot)$  be the (possibly weighted) loss used to define disagreement.

### Where `"gwet"` differs from `"available"`

In `misskappa`, the methods `"available"`, `"ipw"`, and `"gwet"` differ only through the weights used in the nonparametric estimators (raw categorical and continuous backends).

### Observed disagreement $\hat{d}$ (within-subject pairs)

The observed disagreement estimate is computed from within-subject rater pairs:

$$\hat{d} = \frac{\sum_{i=1}^n \sum_{1 \leq j < k \leq R} M_{ij} M_{ik} \ell(X_{ij}, X_{ik}) w_{jk}^{(\text{obs})}}{\sum_{i=1}^n \sum_{1 \leq j < k \leq R} M_{ij} M_{ik} w_{jk}^{(\text{obs})}}.$$

The weight choice is:

$$w_{jk}^{(\text{obs})} = \begin{cases} 1 & \text{method} = \text{"available"} \\ \hat{\pi}_{jk}^{-1} & \text{method} = \text{"ipw"} \\ 1 & \text{method} = \text{"gwet"}. \end{cases}$$

So "gwet" and "available" use the same observed-disagreement component in the current code.

### Chance disagreement $\hat{d}_C, \hat{d}_F$ (cross-subject pairs)

The chance-disagreement components (used for Conger/Fleiss kappas) are based on cross-subject pairings and use product weights:

$$w_{jk}^{(\text{exp})} = \begin{cases} 1 & \text{method} = \text{"available"} \\ \hat{\pi}_j^{-1} \hat{\pi}_k^{-1} & \text{method} = \text{"ipw"} \\ \hat{\pi}_j^{-1} \hat{\pi}_k^{-1} & \text{method} = \text{"gwet"}. \end{cases}$$

Thus, in `misskappa`, "gwet" is *not* the same as "available": it leaves  $\hat{d}$  unchanged but reweights the chance term using inverse *marginal* observation rates (and does *not* use inverse pairwise observation rates in  $\hat{d}$  as "ipw" does).

### Practical consequences

- With complete data ( $M_{ij} \equiv 1$ ),  $\hat{\pi}_j \equiv 1$  and "available" and "gwet" coincide.
- If missingness rates differ by rater (some columns have more NAs), then "gwet" differs from "available" through the chance term.
- For counts data (`kappa_counts()`), "gwet" is not supported because the counts format does not retain per-rater missingness.